

Homework Assignment 1

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A true story: One of the main theorems at a certain PhD thesis defense at the University of Toronto asserted (roughly) that for a certain power series $\sum_{n \geq 1} a(n)z^n$ known to have a positive radius of convergence, the coefficients $a(n)$ satisfied

$$|a(n) + (-1)^n n!| \leq c^n$$

for some constant $c > 1$. A member of the defense committee objected that this theorem couldn't possibly be correct. Why?

We start with a convenient counterexample to the theorem. Take $a(n) = 0$ for all n , and suppose for contradiction that such a c exists. Then the theorem states

$$|(-1)^n n!| \leq c^n.$$

Note that $|(-1)^n n!| = n!$, so the theorem now states that $n! \leq c^n$ for all n and a fixed constant c . However, $n!$ tends to ∞ faster than c^n , so there exists some $N > n$ for which $N! > c^n$, showing that this theorem cannot be true for all n .

Note that this theorem will also fail in the case $a(n) > 0$, since we can choose n even such that $|a(n) + (-1)^n n!| = a(n) + n!$. Since this is now strictly greater than $n!$, the same logic applies. If $a(n) < 0$, we choose n odd and the statement to obtain $|a(n)| + n!$, and the statement once again follows.